

Vraag 1 [4]

Gebruik lus-analise om twee vergelykings van die volgende vorm op te stel:

$$ai_1 + bi_3 = 33$$

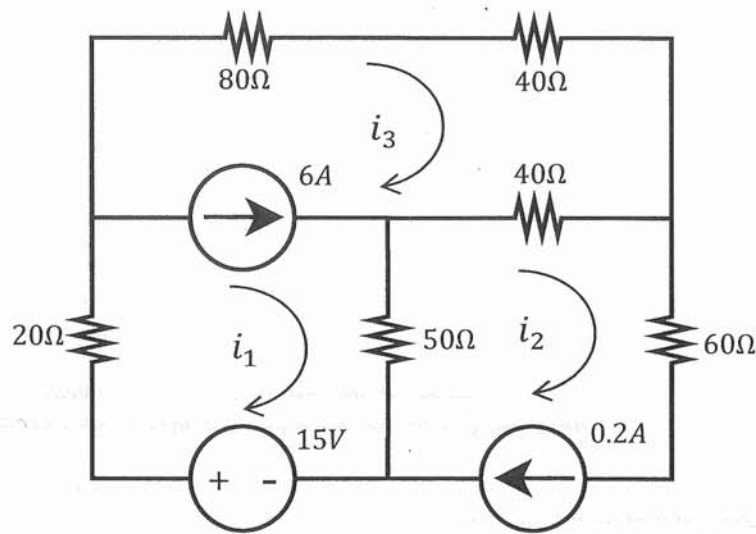
$$ci_1 + di_3 = 6$$

Question 1 [4]

Use mesh analysis to set up two equations of the following form:

$$ai_1 + bi_3 = 33$$

$$ci_1 + di_3 = 6$$



+ Supernesh:

$$0 = -15 + 20i_1 + 80i_3 + 40i_3 + 40i_3 - 40i_2 + 50i_1 - 50i_1$$

$$70i_1 - 40i_2 + 160i_3 = 15$$

$$\text{But } i_2 = 0.2A$$

$$70i_1 + 160i_3 = 15 + 40 \times 0.2 = 33$$

$$70i_1 + 160i_3 = 33$$

* KCL:

$$i_1 = i_3 + 6$$

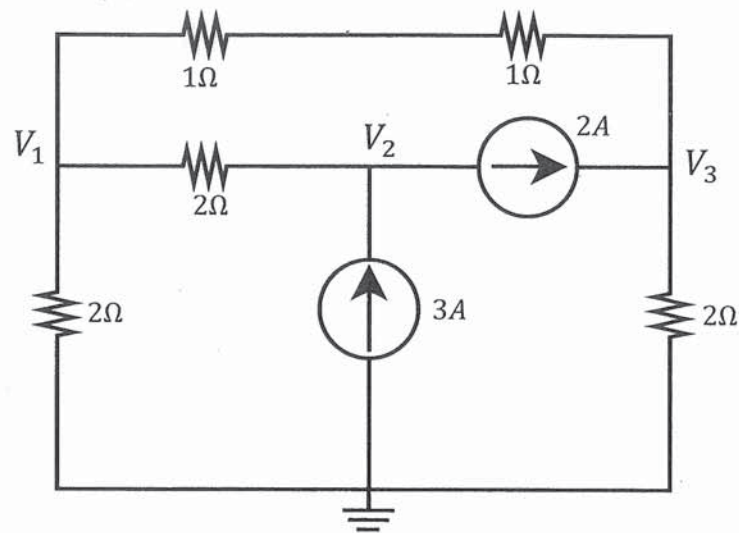
$$i_1 - i_3 = 6$$

Vraag 2 [3]

Stel 'n matriksvergelyking d.m.v. inspeksie op in terme van die nodespannings.

Question 2 [3]

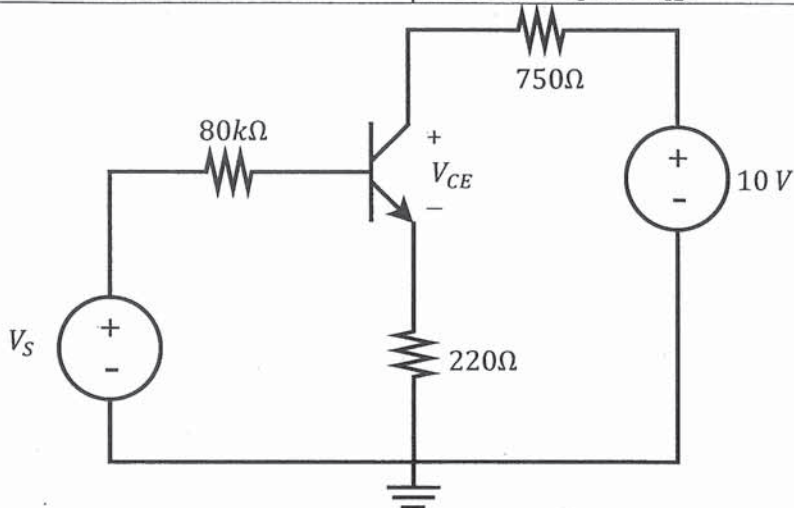
Use inspection to set up a matrix equation in terms of the nodal voltages.



$$\begin{bmatrix} \frac{1}{2} + \frac{1}{2} + \frac{1}{1+1} & -\frac{1}{2} & -\frac{1}{1+1} \\ -\frac{1}{2} & \frac{1}{2} & 0 \\ -\frac{1}{1+1} & 0 & \frac{1}{2} + \frac{1}{1+1} \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 3-2 \\ 2 \end{bmatrix}$$

$$\begin{bmatrix} 1,5 & -0,5 & -0,5 \\ -0,5 & 0,5 & 0 \\ -0,5 & 0 & 1 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}$$

✓ → Diagonal
 ✓ → Off-diagonal
 ✓ → Result

Vraag 3 [4]Gegee: $I_B = 0.08 \text{ mA}$, $\beta = 99$ en $V_{BE} = 0.7 \text{ V}$.Bepaal V_S en V_{CE} .**Question 3 [4]**Given: $I_B = 0.08 \text{ mA}$, $\beta = 99$ and $V_{BE} = 0.7 \text{ V}$.Determine V_S and V_{CE} .

$$\begin{aligned}
 \textcircled{1} \quad & -V_S + 80k\Omega \cdot I_B + 0.7 + 220 \times I_E = 0 \\
 & V_S = 80 \times 10^3 \times I_B + 0.7 + 220 (\beta + 1) I_B = 0 \\
 & V_S = 0.7 + [80 \times 10^3 + 220 (99 + 1)] \times 0.08 \times 10^{-3} \\
 & V_S = \underline{8.86 \text{ V}} \quad \checkmark
 \end{aligned}$$

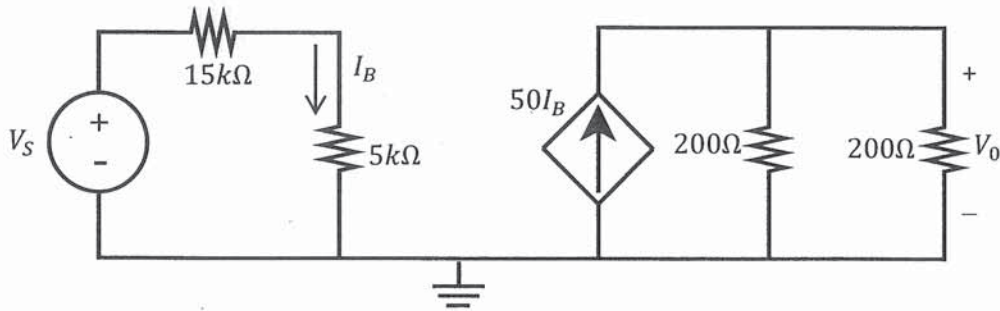
$$\begin{aligned}
 \textcircled{2} \quad & -10 + 750 \times I_C + V_{CE} + 220 I_E = 0 \\
 & V_{CE} = 10 - 750 (99) I_B - 220 (100) I_B \\
 & = \underline{2.3 \text{ V}} \quad \checkmark
 \end{aligned}$$

Vraag 4 [2]

Maak gebruik van die skalingseienskap van lineariteit om die verhouding V_0/V_S te bepaal.

Question 4 [2]

Make use of the scaling property of linearity to determine the relationship V_0/V_S .



* Assume $V_0 = 1 \text{ V}$.

$$\rightarrow \therefore 50 I_B = \frac{1}{200} + \frac{1}{200}$$

$$I_B = \left(\frac{2}{200} \right) / 50 = 0,2 \text{ mA}$$

$$\rightarrow \therefore -V_S + 15 \text{ k}\Omega I_B + 5 \text{ k}\Omega I_B = 0$$

$$V_S = 20 \times 10^3 \times 0,2 \times 10^{-3}$$

$$= \underline{4 \text{ V}}$$

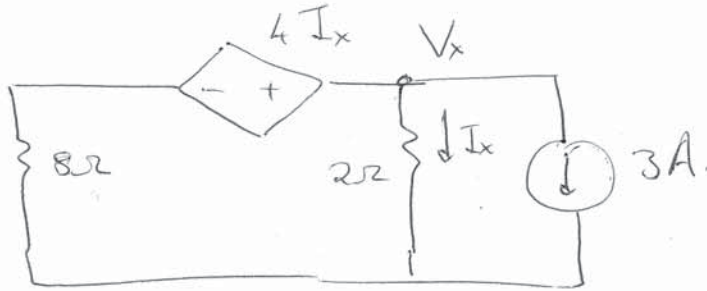
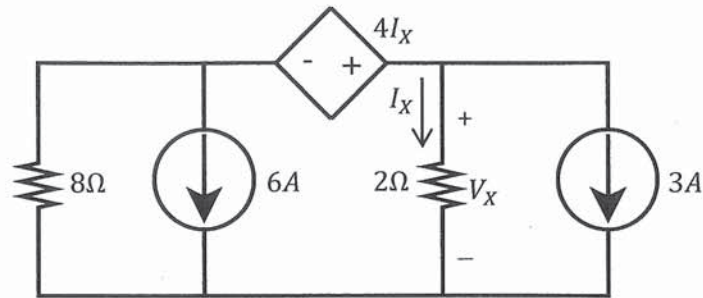
$$\therefore \frac{V_0}{V_S} = \frac{1}{4} = \underline{0,25} \quad \checkmark \checkmark$$

Vraag 5 [2]

Bepaal die bydrae tot V_X slegs as gevolg van die 3A stroombron.

Question 5 [2]

Determine the contribution to V_X by the 3A current source only.



$$* \quad I_X = \frac{V_X}{2}$$

$$\text{KCL: } * \quad 3 + \frac{V_X}{2} + \frac{V_X - 4I_X}{8} = 0$$

$$(\text{Sub:}) \quad 3 + \frac{V_X}{2} + \frac{V_X - 4\left(\frac{V_X}{2}\right)}{8} = 0$$

$$3 + \frac{V_X}{2} + \frac{V_X - 2V_X}{8} = 0$$

$$3 + \frac{V_X}{2} - \frac{V_X}{8} = 0$$

$$8 \times 3 = V_X - 4V_X$$

$$8 \times 3 = -3V_X$$

$$V_X = -8V \quad \checkmark$$